Trauma for the Shelf: Every Topic in 45 Minutes

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Eight Objectives Across Ten Trauma Domains

1. Sequence xABCDE — intervene before imaging
2. Classify shock I-IV — MTP by Shock Index
3. Distinguish epidural vs subdural by CT shape
4. Identify cord syndromes by deficit pattern
5. Calculate Parkland from burn time
6. Differentiate tension PTX vs tamponade at bedside
7. Apply FAST — blunt vs penetrating algorithms
8. Recognize DCS triggers + peds/geri/pregnancy mods

Every Trauma Patient Gets the Same Systematic Assessment — ABCDE

- A = Airway (with c-spine protection)
- B = Breathing (chest injuries)
- C = Circulation (hemorrhage, shock, FAST)
- D = Disability (GCS, pupils, neuro exam)
- E = Exposure (full exam, prevent hypothermia)
- Taught worldwide since 1980 — the foundation of ATLS

But What If the Patient Is Bleeding to Death Before You Even Assess the Airway?

- Eastridge studied 4,596 battlefield deaths (2001-2011)
- Which was the #1 cause of preventable death?
- Airway obstruction... or hemorrhage?

Hemorrhage Kills First — ATLS 11th Edition Added X Before A

- xABCDE → hemorrhage first (ATLS 11th Ed, 2025)
- Eastridge: 90.9% of preventable deaths = bleeding
- X step: tourniquet, pressure, binder, MTP
- Then A → B → C → D → E in sequence
- What kills this patient first — the airway or the hemorrhage?

Tourniquet 15 sec vs RSI 3-5 min — the time gap kills

Unstable patient offered CT — answer is intervention first, not imaging

Class III Is the Inflection Point — That's When You Give Blood

Class I	Class II	Class III	Class IV
Blood loss	<15%	15-30%	30-40%
HR	<100	100-120	>120
BP	Normal	Normal	↓
Rx	Monitor	Crystalloid	Blood

BP doesn't drop until 30% is lost — young patients compensate then crash

BP stays normal until Class III — don't trust it

HR 130, BP 80 with trauma — answer is blood products, not 2L normal saline

Shock Index Tells You What the Blood Pressure Won't

- Shock Index = HR ÷ SBP (normal 0.5-0.7)
- SI >1.0 = significant hemorrhage
- SI >1.4 = activate MTP
- PROPPR trial: 1:1:1 ratio (pRBC:FFP:platelets)
- How fast can you calculate your patient's Shock Index right now?

SI >1.0 = 4x mortality

Shock Index >1.0 with hemorrhage — answer is MTP activation, not crystalloid bolus

TXA Has a Hard 3-Hour Cutoff — Before It Saves, After It Kills

- CRASH-2: largest trauma RCT ever (n=20,211)
- <1 hour: 32% reduction in hemorrhagic death
- 1-3 hours: 21% reduction
- >3 hours: 44% INCREASE — actively harmful
- Dose: 1g IV bolus then 1g over 8h

3 hours from injury — not from arrival

2 hours post-injury — give TXA now; 4 hours post — answer is withhold

Clinical Decision Point

32M, GSW left chest and abdomen. HR 140, BP 70/40, GCS 7. Active abdominal hemorrhage, gurgling respirations, no breath sounds on left.
First step?

A. Intubate — GCS 7 means secure the airway (A step)

B. Direct pressure + activate MTP — control hemorrhage (X step)

C. Needle decompression — absent breath sounds (B step)

D. CT chest/abdomen/pelvis — define the injuries

This tests xABCDE priority. All four letters present: X (active hemorrhage), A (GCS 7), B (absent BS), and C (shock). Each answer maps to a different step. Give them 30 seconds. Which letter comes first?

> Shelf Tip: When competing life threats coexist, follow xABCDE alphabetically. Earliest letter =

Answer: B — Hemorrhage Control First, Because X Comes Before A

Why B: active exsanguination = X step, first priority

Why not A: GCS 7 needs intubation, but X before A

Why not C: absent BS = B step, follows X and A

Why not D: CT in unstable patient = never correct

Tourniquet 15 sec vs RSI 3-5 min — the time gap kills

Active hemorrhage + low GCS — answer is hemorrhage control, not intubation

TBI Is the #1 Cause of Trauma Death — And Most of It Is Preventable

- 61,000 Americans die from TBI annually
- Leading cause of trauma death under age 45
- Primary injury = the impact (can't change it)
- Secondary injury = hypotension, hypoxia (CAN prevent it)
- What is the single most important thing you can control after TBI?

SBP <90 doubles TBI mortality

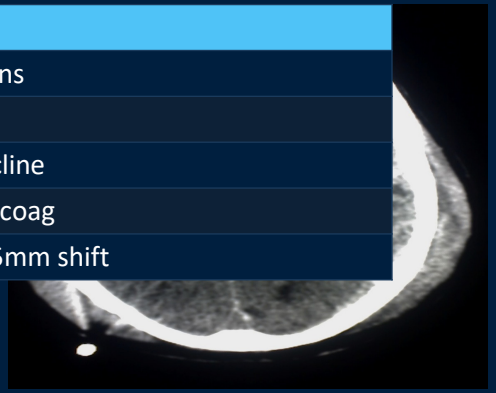
TBI patient with low BP — answer is raise MAP, not hyperventilate

Two CT Scans, Two Different Emergencies — Can You Tell Them Apart?

- One is epidural. One is subdural.
- Different source, different patient, different prognosis
- Which shape is which?

Biconvex = Epidural, Crescent = Subdural — Everything Else Follows

Feature	Epidural	Subdural
Source	MMA (arterial)	Bridging veins
CT shape	Biconvex	Crescent
History	Lucid interval	Gradual decline
Population	Young, temporal fx	Elderly, anticoag
OR threshold	>1cm / >30mL / shift	>10mm / >5mm shift



Epidural with early evacuation = good prognosis

CT shows biconvex hyperdensity — answer is epidural, not subdural

Three Herniation Patterns — Each Has One Finding You Cannot Miss

- Uncal → CN III → ipsilateral blown pupil → contralateral hemiparesis
- Central → bilateral fixed pupils → decorticate then decerebrate
- Cushing triad → HTN + bradycardia + irregular respirations
- If you see Cushing triad — how much time do you have?

The picture can't be displayed.

Cushing triad = minutes to brainstem death

Ipsilateral fixed dilated pupil — answer is uncal herniation, not central

ICP Management Escalates in Three Tiers — Know When to Move Up

Tier	What you do	Target
1	HOB 30°, sedation, CSF drain	ICP <22
2	Hypertonic saline / mannitol	CPP 60-70
3	Craniectomy, barbiturate coma	Refractory

Tier 1 fails → Tier 2 (osmotherapy) → Tier 3 (surgery)
"More mannitol" is never the answer for refractory ICP

ICP threshold = 22 mmHg (updated from 20 in BTF 4th Ed)

Refractory ICP despite osmotherapy — answer is decompressive craniectomy, not more mannitol

Clinical Decision Point

28M found after assault with baseball bat to right temple. Initial GCS 15 — talked to the nurse. 30 minutes later: GCS 6, right pupil fixed and dilated. CT shows biconvex hyperdensity with 8mm midline shift.
Next step?

A. IV hypertonic saline and repeat CT in 1 hour

B. HOB 30°, sedate, place ICP monitor

C. Emergent craniotomy for hematoma evacuation

D. Hyperventilate to PaCO2 25 and call neurosurgery

Classic epidural. Lucid interval — talked to nurse, then rapid decline. Biconvex CT with shift. Blown pupil = herniation NOW. Which tier of management — or does this patient need something beyond tiers?

> Shelf Tip: Biconvex + shift + declining GCS = surgical emergency, not ICP monitoring.

Answer: C — Biconvex + Shift + Blown Pupil = OR, Not Monitoring

Why C: epidural with herniation = emergent surgical evacuation

Why not A: osmotherapy is a bridge, not definitive for surgical lesion

Why not B: ICP monitor for diffuse injury, not a shifting mass

Why not D: PaCO2 25 causes ischemia — bridge only, target 30-35

Epidural prognosis is good with early evacuation — delay = death

Biconvex CT + midline shift + declining GCS — answer is craniotomy, not ICP monitor

Four Cord Syndromes — Each Has a Signature Deficit Pattern

- Same spinal cord, four different injury patterns
- Each produces a unique motor/sensory combination
- If I tell you "arms are weaker than legs" — which syndrome?
- The answer depends on the patient's age and mechanism

Central cord is the most common incomplete SCI

Arms worse than legs in elderly after fall — answer is central cord, not anterior

Match the Deficit to the Syndrome — The Pattern Is the Diagnosis

Syndrome	Mechanism	Lost	Preserved
Anterior	Burst fx / aortic	Motor, pain, temp	Proprio, vibration
Central	Hyperextension (elderly)	Arms >> legs	Sacral sensation
Brown-Sequard	Hemisection (penetrating)	Ipsi motor; contra pain/temp	Opposite modalities
Cauda equina	L1-L2 compression	Saddle, bowel/bladder	Surgical emergency

Central cord = most common incomplete SCI

Arms worse than legs in elderly — answer is central cord, not anterior

Two Rules Clear the C-Spine — CCR Is More Specific

NEXUS	CCR
Sensitivity	99.6%
Specificity	12.9%
Criteria	5 low-risk items
Best for	Universal, quick
Obtunded	CT alone (NPV >99%)

Both near-perfect sensitivity — the difference is unnecessary CTs

CCR is 3.5x more specific — fewer patients irradiated

Alert stable trauma with neck pain — answer is apply CCR, not automatic CT

Neurogenic Shock Looks Like the Opposite of Everything You Learned About Shock

- Hemorrhagic: tachy + cold + vasoconstricted
- Neurogenic: brady + warm + vasodilated
- Loss of sympathetic tone below T6 = unopposed vagal
- Rx: norepinephrine first-line, MAP >85 × 7 days
- Your patient has HR 52 and BP 78/50 after a C5 fracture — is this hemorrhagic?

Bradycardia is the discriminator

Hypotension + bradycardia + warm — answer is neurogenic shock, not hypovolemic

Clinical Decision Point

22F diving accident into shallow pool. C5 burst fracture. BP 78/50, HR 52, warm dry extremities. No external bleeding, abdomen soft, FAST negative.
Best initial vasopressor?

A. Phenylephrine (pure alpha-1 vasoconstriction)

B. Norepinephrine (alpha-1 + beta-1)

C. Dopamine (dose-dependent effects)

D. Vasopressin (V1 receptor agonist)

Classic neurogenic shock. C5 fracture, bradycardia, hypotension, warm extremities, FAST negative rules out hemorrhage. Each answer is a different vasopressor. The question tests WHY norepinephrine is first-line.

> Shelf Tip: If bradycardic with spinal injury, pick the vasopressor that treats BOTH vasodilation

Answer: B — Norepinephrine Treats Both Vasodilation and Bradycardia

Why B: alpha-1 (vasoconstriction) + beta-1 (↑ HR) = both problems

Why not A: phenylephrine = pure alpha, no beta — worsens bradycardia

Why not C: dopamine is second-line, less predictable

Why not D: vasopressin has no chronotropic effect — doesn't fix HR

NE = the only first-line that addresses both components

C-spine injury + bradycardia + hypotension — answer is norepinephrine, not phenylephrine

Burns Are Classified by Depth — And the Most Dangerous Ones Don't Hurt

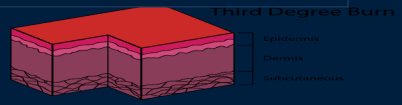
- Superficial = painful (nerves intact)
- Partial thickness = very painful (most painful of all)
- Full thickness = painless (nerves destroyed)
- A patient with a charred, painless wound — is that mild or severe?

Painless burn = full thickness = the classic shelf trick

Charred painless leathery wound — answer is full thickness, not deep partial

Burn Depth Determines Sensation, Healing, and Whether You Need a Graft

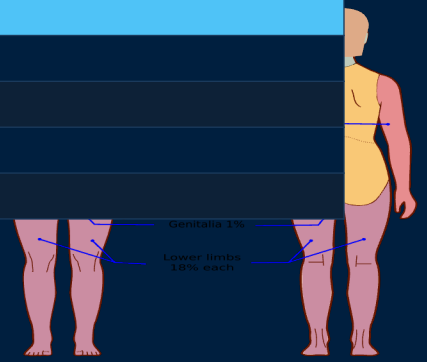
Depth	Appearance	Sensation	Healing
Superficial	Red, dry	Painful	3-5 days
Partial	Pink, blisters	Very painful	2-3 weeks
Deep partial	Mottled	Reduced	3-8 wk, scars
Full thickness	Leathery, white/brown	Painless	Graft required
Subdermal	Charred	Painless	Excision



Painless + leathery = full thickness = requires grafting, not conservative management

Rule of Nines Estimates TBSA — Parkland Starts the Clock at Burn Time

Region	Adult	Pediatric
Head	9%	18%
Each arm	9%	9%
Trunk (ant/post)	18% / 18%	18% / 18%
Each leg	18%	14%



Parkland: 4 × kg × %TBSA = 24h total (LR only)

Half in first 8 hours from BURN time — not arrival

The clock starts when the patient burns, not when they arrive

Calculate from 1400 burn time, not 1600 arrival — most common shelf error

Inhalation Injury Closes the Airway in Hours — Intubate Before It's Too Late

- Suspect: enclosed fire + singed hairs + carbonaceous sputum
- CO poisoning: 100% O2 (half-life 250 → 40 min)
- Cyanide: refractory lactic acidosis → hydroxocobalamin
- Escharotomy: circumferential burn + absent pulses (NOT fasciotomy)

Airway edema progresses over hours — the window closes fast

Enclosed fire + carbonaceous sputum — answer is intubate now, not observe

Clinical Decision Point

80 kg man, 40% TBSA flame burn at 1400h. Arrives ED at 1600h.
Parkland formula with LR — how much fluid between 1600h and 2200h?

A. 3,840 mL

B. 5,120 mL

C. 6,400 mL

D. 12,800 mL

Classic Parkland calculation. The trap: calculating from arrival time. The 8-hour clock started at 1400 when the burn happened. They arrive at 1600, so 2 of 8 hours are gone. Work through: total, first-8-hour half, subtract elapsed time. Give them 60 seconds with scratch paper.

> Shelf Tip: Draw the timeline. Burn 1400, arrive 1600, 8h mark 2200.

Answer: B — 5,120 mL Because the Clock Started at Burn Time

Total 24h: 4 × 80 × 40 = 12,800 mL

First 8h (1400-2200): 6,400 mL

Already elapsed (1400-1600): ~1,280 mL owed

Remaining (1600-2200): 6,400 − 1,280 = 5,120 mL in 6 hours

12,800 mL

Parkland from burn time, not arrival — the #1 tested calculation error

Most Chest Trauma Deaths Are Treatable With a Chest Tube — Know the Thresholds

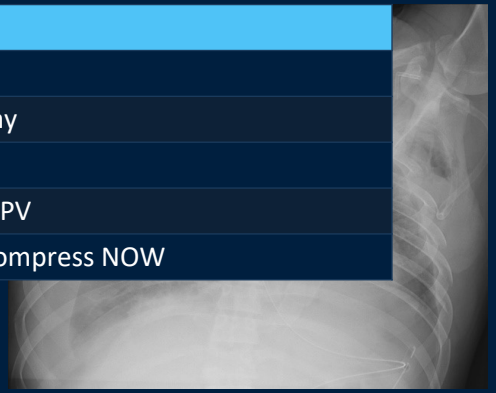
- 25% of trauma deaths involve chest injury
- 85% managed with chest tube alone — no thoracotomy
- Two numbers to memorize for the shelf...

1,500 mL initial — or 200 mL/hr × 2-4h — = thoracotomy

Chest tube drains 1,800 mL immediately — answer is thoracotomy, not observe

Each Chest Condition Has a Specific Action Trigger

Condition	Threshold	Action
Simple hemothorax	<1,500 mL	Chest tube
Massive hemothorax	≥1,500 mL or >200/hr	Thoracotomy
Retained hemothorax	Persistent post-tube	VATS 72h
Flail chest	≥3 ribs, ≥2 places	Epidural + PPV
Tension PTX	Clinical diagnosis	Needle decompress NOW



Tension PTX = clinical diagnosis — never wait for CXR; ≥1,500 mL = OR

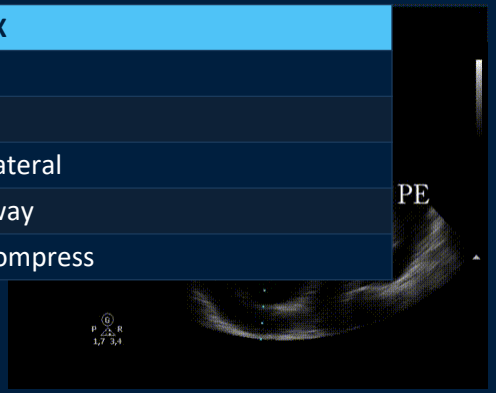
Widened Mediastinum After Deceleration = CTA for Aortic Injury

- Mechanism: rapid deceleration (MVC >40 mph, fall >15 ft)
- Tears at the aortic isthmus (distal to L subclavian)
- CXR clue: mediastinum >8 cm, left apical cap
- CTA is gold standard — endovascular repair preferred

High-speed MVC + wide mediastinum — answer is CTA, not repeat CXR

Both Cause JVD + Hypotension — Breath Sounds Tell You Which One

Finding	Tamponade	Tension PTX
JVD	Yes	Yes
Hypotension	Yes	Yes
Breath sounds	Bilateral (normal)	Absent unilateral
Trachea	Midline	Deviated away
Treatment	Pericardiocentesis	Needle decompress



EDT: penetrating cardiac ~15% survival; blunt ~1.5%

Breath sounds = the single discriminator

JVD + hypotension + bilateral BS = tamponade; absent unilateral BS = tension PTX

Clinical Decision Point

28F, stab wound to left chest. HR 125, BP 75/45, SpO2 90%. Absent breath sounds on the left, JVD present, trachea deviated right.
Most appropriate first step?

A. Pericardiocentesis (JVD + hypotension = tamponade?)

B. Left needle decompression at 2nd ICS

C. FAST exam to evaluate for tamponade

D. Emergent left thoracotomy

The stab wound is a deliberate distractor — makes you think tamponade. But read the exam: absent LEFT breath sounds, tracheal deviation RIGHT. That's tension PTX. Tamponade has bilateral BS and midline trachea. Give them 30 seconds.

> Shelf Tip: Mechanism suggests injury, physical exam makes the diagnosis. Trust the exam

Answer: B — Absent Breath Sounds + Tracheal Deviation = Tension PTX, Not Tamponade

Why B: absent left BS + rightward tracheal deviation = tension PTX

Why not A: tamponade has bilateral BS + midline trachea — doesn't fit

Why not C: tension PTX = clinical diagnosis — FAST delays treatment

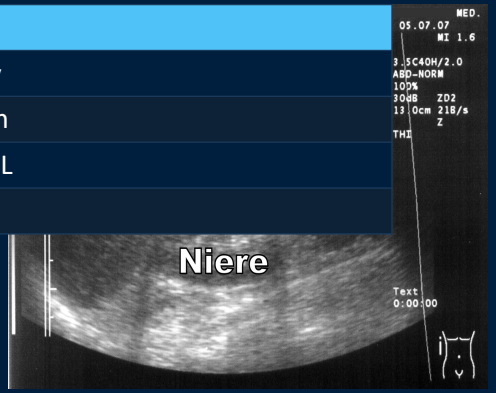
Why not D: EDT for arrest/peri-arrest — this patient has a pulse

The stab wound was the distractor — the physical exam is the diagnosis

Absent unilateral BS + tracheal deviation = needle decompress NOW, not image

FAST Positive + Unstable = OR — No CT, No DPL, No Delay

FAST	Hemodynamics	Action
Positive	Unstable	Laparotomy
Positive	Stable	CT abdomen
Negative	Unstable	Repeat / DPL
Negative	Stable	Observe



Spleen NOM: 85% (grade I-III) — vaccinate post-splenectomy
Liver NOM: 92% (grade I-III)
FAST sens 73%, spec 96% — rules in, does NOT rule out

Positive FAST + hypotension — answer is laparotomy, not CT scanner

GSW to Abdomen = OR. Stab Wound = It Depends.

- GSW abdomen → exploratory laparotomy (nearly always)
- Stab wound + unstable → OR
- Stab wound + stable → local wound exploration
- Fascia violated → CT or diagnostic laparoscopy
- Evisceration → OR regardless of stability
- A patient walks in with a stab wound and normal vitals — do you operate?

Gunshot wound to abdomen — answer is exploratory laparotomy, not CT

Two Zone Systems Every Surgeon Must Know — Neck and Retroperitoneum

Neck Zone	Landmarks	Management
I	Clavicles → cricoid	CTA
II	Cricoid → mandible angle	Selective CTA
III	Above mandible	Angiography / IR
Hard signs	Any zone	Immediate OR

Hard signs any neck zone = OR; Zone 3 pelvic hematoma at laparotomy = leave it alone

Clinical Decision Point

22M restrained driver, MVC. FAST positive in Morison's pouch. BP 70/40 despite 2 units pRBCs. GCS 14. Next step?

A. CT abdomen/pelvis with IV contrast

B. Diagnostic peritoneal lavage

C. Emergent exploratory laparotomy

D. Repeat FAST in 30 minutes

FAST positive, unstable despite blood products. Top-left cell of the 2×2: FAST+ unstable = OR. The temptation is CT or DPL. But this patient has declared himself: free fluid + refractory hypotension = surgical hemorrhage.

> Shelf Tip: BP still low despite blood + FAST positive = OR. No more tests.

Answer: C — FAST Positive + Refractory Hypotension = Laparotomy

Why C: FAST+ with hypotension despite 2u pRBC = surgical hemorrhage

Why not A: CT requires stability — BP 70/40 = unstable

Why not B: DPL for equivocal FAST — this is clearly positive

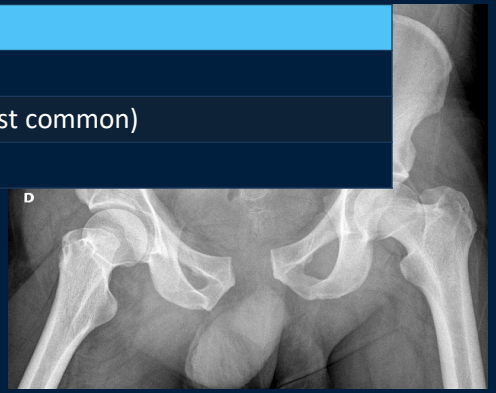
Why not D: diagnosis is made — repeating delays treatment

FAST+ unstable = OR — the CT scanner is not in the algorithm

Positive FAST + refractory hypotension — answer is laparotomy, not CT or DPL

Pelvic Fractures: Binder First, FAST Second, Then Decide OR vs Angio

Type	Mechanism	Mortality
APC (open book)	Head-on MVC	Moderate
LC (lateral compression)	Side-impact	Lowest (most common)
VS (vertical shear)	Fall from height	Highest



Binder at trochanters → FAST → decide
FAST+ = laparotomy first (not pelvic packing)
FAST- unstable = angioembolization
Blood at meatus = RUG before Foley — no exceptions

Unstable pelvis + positive FAST — answer is laparotomy, not angioembolization

Compartment Syndrome Has a 6-Hour Window — Don't Wait for Pulselessness

- Most common: tibial shaft fracture (anterior compartment)
- Earliest sign: pain with passive stretch
- Latest sign: pulselessness — if you wait for this, it's too late
- Pressure >30 mmHg or delta <30 from diastolic
- Rx: emergent 4-compartment fasciotomy
- A patient with a tibial fracture and increasing pain despite morphine — what are you worried about?

pain with passive stretch

Pain with passive stretch after tibial fracture — answer is fasciotomy, not more pain meds

Open Fracture Grade Drives Antibiotics — IIIC Means Vascular Emergency

Gustilo	Wound	Antibiotics
I	<1 cm, clean	Cefazolin
II	1-10 cm	Cefazolin
IIIA	>10 cm, covered	+ aminoglycoside
IIIB	>10 cm, uncovered	+ flap coverage
IIIC	Vascular injury	+ vascular repair

All open fractures: tetanus + antibiotics within 1 hour + debridement

Fat embolism: 24-72h post long bone fx → dyspnea + confusion + petechiae

IIIC = vascular injury = highest amputation rate

Open fracture with absent distal pulses — answer is Gustilo IIIC, needs vascular repair

Clinical Decision Point

45M motorcycle crash. Open left tibial fracture (Gustilo II), tense painful anterior compartment. HR 118, BP 92/60. Pelvic binder in place. FAST positive in Morison's pouch.
Priority?

A. Emergent 4-compartment fasciotomy

B. Emergent exploratory laparotomy

C. Pelvic angioembolization

D. Tibial fracture washout and external fixation

Multiple problems: FAST-positive abdomen (life threat), compartment syndrome (limb threat), pelvic injury (binder in place). Which first? Life before limb. Compartment has 6 hours. Hemorrhage has minutes. Give them 30 seconds.

> Shelf Tip: Life-threatening vs limb-threatening — always treat life threat first.

Answer: B — Life Before Limb. Hemorrhage Kills in Minutes.

Why B: FAST+ with hemodynamic instability = life-threatening hemorrhage

Why not A: compartment = limb-threat (6h window) — do after laparotomy

Why not C: FAST+ = abdominal source identified, not pelvic

Why not D: fracture management not emergent when unstable

Life before limb — fasciotomy in the same OR after hemorrhage control

Unstable + FAST positive + compartment syndrome — answer is laparotomy first, fasciotomy after

When the Lethal Triad Is Present — Pack, Staple, Close, Resuscitate

- Lethal triad: hypothermia + acidosis + coagulopathy
- Trigger: ≥2 of temp <35 / pH <7.2 / coagulopathy
- Phase 1 (OR <60 min): control hemorrhage, temp closure
- Phase 2 (ICU): rewarm, correct coagulopathy and acidosis
- Phase 3 (24-72h): return to OR for definitive repair

Lethal triad present = damage control, NOT definitive repair

Temp 34, pH 7.18, INR 2.1 at laparotomy — answer is damage control, not definitive repair

Three Populations That Break the Standard Algorithms

Pediatric	Geriatric	Pregnancy
#1 mechanism	MVC	Falls
Key pearl	Rib fx = abuse	Beta-blockers mask tachy
Pathognomonic	Retinal hemorrhages = NAT	Chronic subdural wks post-fall
Resuscitation	20 mL/kg boluses	Lower threshold for everything
Critical action	Skeletal survey	Low threshold CT on anticoag

Infant + bilateral subdurals + retinal hemorrhages — answer is NAT, not accidental fall

Clinical Decision Point

8-month-old male brought by mother's boyfriend. "He rolled off the couch." CT: bilateral subdural hematomas of different ages. Fundoscopy: bilateral retinal hemorrhages.
Most likely diagnosis?

A. Accidental fall consistent with reported mechanism

B. Glutaric aciduria type I (metabolic cause of subdurals)

C. Non-accidental trauma

D. Birth-related subdural hematomas

Classic NAT. Mechanism is inconsistent — couch fall can't produce bilateral subdurals of different ages. Retinal hemorrhages in >80% of abusive head trauma. Glutaric aciduria mimics subdurals but does NOT cause retinal hemorrhages. Birth subdurals resolve by 4-6 weeks. Give them 30 seconds.

> Shelf Tip: Inconsistent mechanism + subdurals + retinal hemorrhages = abuse. Order skeletal

Answer: C — Bilateral Subdurals + Retinal Hemorrhages + Inconsistent History = NAT

Why C: mechanism can't explain bilateral subdurals of different ages + retinal hemorrhages

Why not A: couch fall → linear skull fracture at most, not bilateral subdurals

Why not B: glutaric aciduria → subdurals but NOT retinal hemorrhages

Why not D: birth subdurals resolve by 4-6 weeks — this is 8 months old

Retinal hemorrhages present in >80% of abusive head trauma

Infant + bilateral subdurals + retinal hemorrhages + inconsistent history — answer is NAT, order skeletal survey

Ten Topics, Ten Discriminators — What to Remember Walking Into the Exam

1. xABCDE: unstable + CT offered = intervention first
2. Shock: SI >1.0 = blood products, not crystalloid
3. TBI: biconvex = epidural; SBP <90 = 2x mortality
4. Spine: arms > legs = central cord; brady = neurogenic
5. Burns: painless = full thickness; Parkland from burn time
6. Chest: 1,500 / 200 = thoracotomy; BS tells tamponade vs PTX
7. Abdomen: GSW = OR; FAST+ unstable = OR
8. Pelvis: binder → FAST → decide; meatus = RUG before Foley
9. Ortho: passive stretch = fasciotomy; IIIC = vascular repair
10. Special: rib fx child = abuse; beta-blockers mask hemorrhage

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